

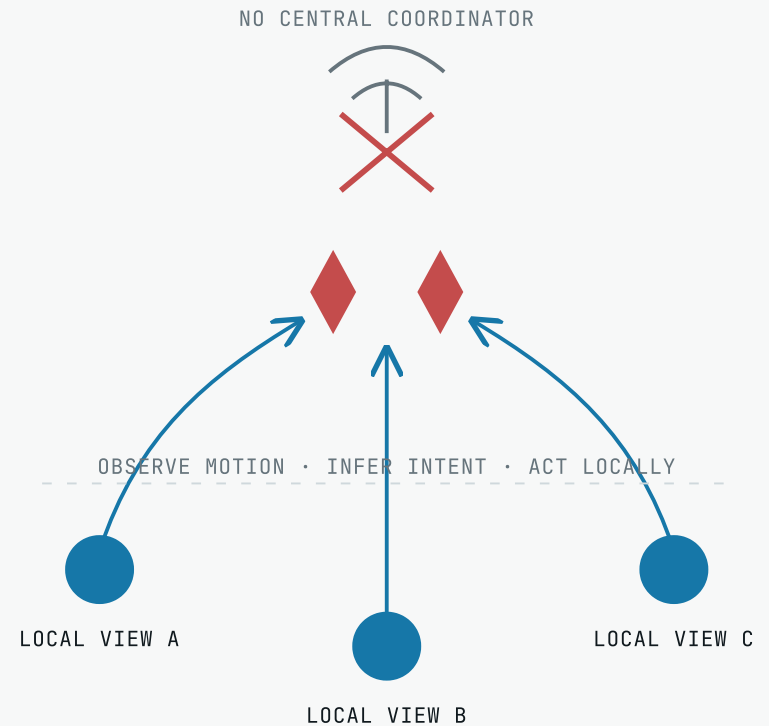
HUSH / AirDnD

HUSH concept / AirDnD simulation

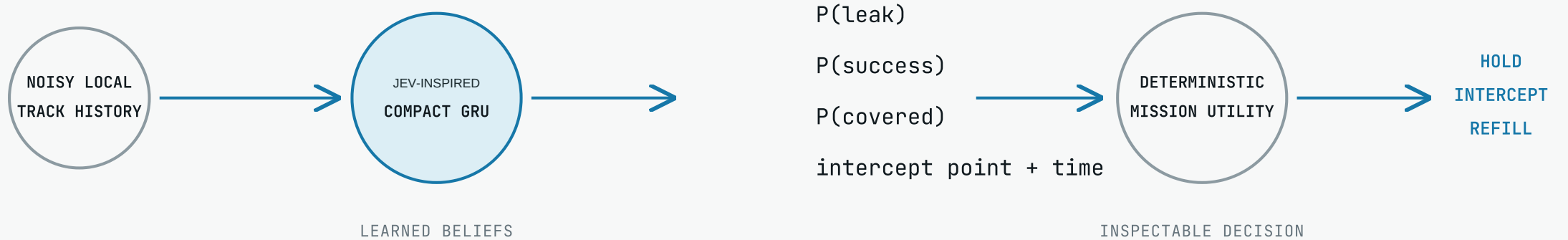
Coordination without shared truth.

When GNSS and every RF link are gone, each interceptor must decide alone—without acting alone.

ZERO GROUND LINKS • ZERO INTER-DRONE MESSAGES • ZERO ASSIGNMENT MESSAGES



Every interceptor runs the same loop



$$U_{ij} = C_j \cdot P(L_j) \cdot P(S_{ij}) \cdot [1 - P(C_j)] - C_E - C_B - C_G - C_R$$

Risk

consequence × leak

Ability

my success

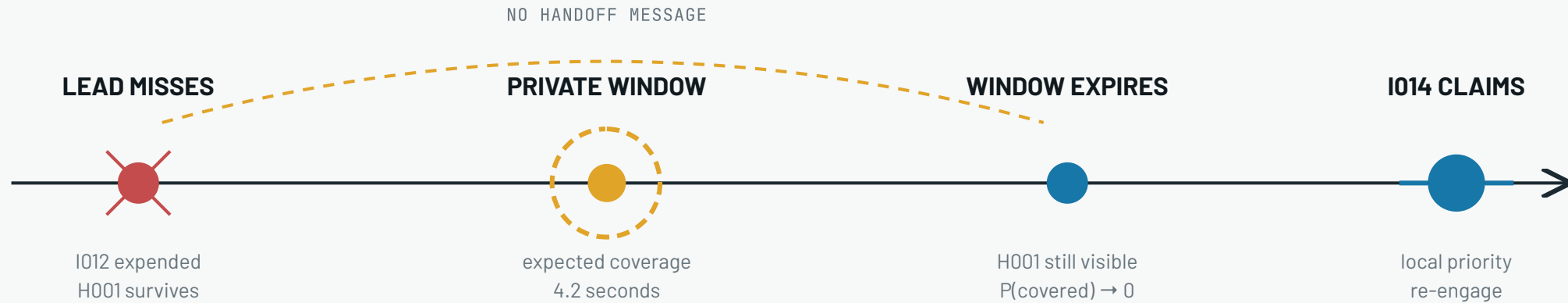
Redundancy

friendly coverage

Costs

round · battery · grid · collision

A miss becomes a local signal

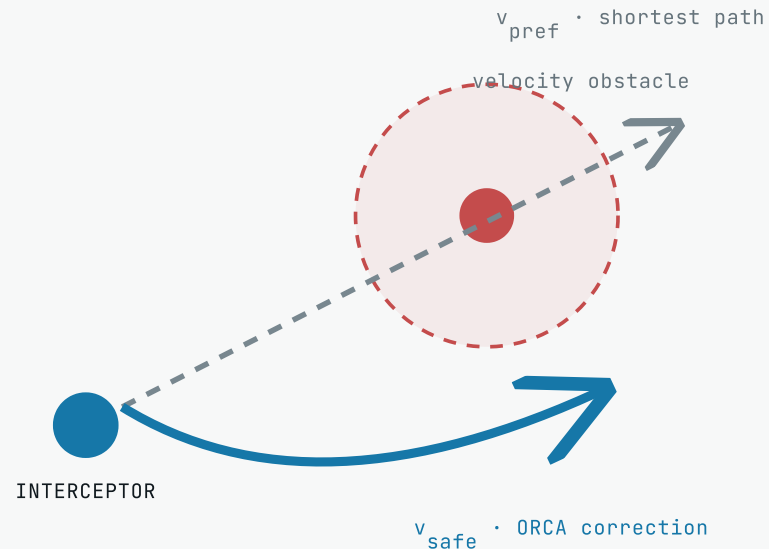


CLAIM PRIORITY

sector → utility → success → intercept time → grid loss → battery → stable track ID

$$t_{\text{claim}} = t_{\text{expiry}} + \text{rank} \cdot \delta, \quad \delta = 0.15 \text{ s}$$

Guidance proposes. Safety can veto.



1 · Predict the basket

$$v_{pref} = \lim_{x \rightarrow x_{int}} (x_{int} - x) / \max(\tau, \tau_{min})$$

Receding-horizon guidance aims where the target will be.

2 · Remove collision velocity

$$v_{cmd} = ORCA(v_{pref}, \text{local neighbours})$$

ORCA returns the nearest locally collision-free command.

3 · Terminal guidance

$$a_{PN} = N \cdot V_c \cdot \dot{\lambda}$$

Proportional navigation drives line-of-sight rate toward zero.

Tested at raid scale

100

hostiles per run

125

interceptors

30

paired seeds

-28.8 percentage points

leakage versus independent greedy

95% CI: 27.4-30.2 points · 53.4% relative reduction

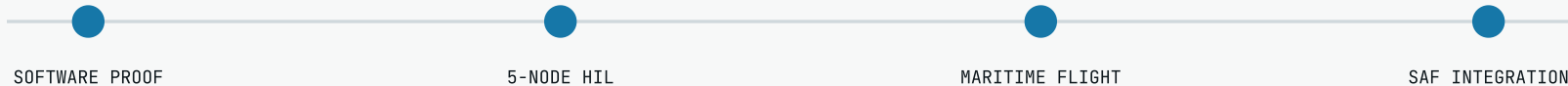
0 friendly collisions

0 entity drops

19.04 ms p95 normalised per interceptor

20 KB INT8 model

Timing = total scenario elapsed time / 125 interceptors



Apple M3 simulation evidence / fixed configuration / archived seeds and raw logs

Roadmap stages are proposed; integration is subject to validation and partner agreement.